gpeyt1-31 · Corrosion allowance

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

Uncoated carbon-steel H-piles will be driven through a fill with field resistivity 1,500 Ω·cm (pH 5.8, sulfates 400 ppm) for a bridge with a 75-year design life. The owner requires the maximum (conservative) loss rate per face. Flange thickness is 0.61 in.

Most nearly, the sacrificial thickness loss per exposed face over the design life is:

A. 0.099 in.
B. 0.049 in.
C. 0.016 in.
D. 0.187 in.

Given equations / data

> x = K tⁿ (μm); carbon steel n = 0.80, K = 40 (avg) / 80 (max); 1,000 μm = 1 mm = 0.039 in.

Worked solution

1. 1,500 Ω·cm → corrosive (700–2,000) per resistivity table
2. x = 80 × 75^0.80 = 2,530 μm
3. = 2.53 mm × 0.039 = 0.099 in. per face

Final answer

0.099 in.

Why the other values are wrong

- **0.049 in.** — used the average K = 40
- **0.016 in.** — used galvanized-steel constants
- **0.187 in.** — multiplied t by n instead of raising to n

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.17.1 Reactive and Corrosive Soils (resistivity table), p. 197; §3.17.2 Corrosion of Buried Steel, p. 198

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-32 · Rock slope with joint water

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A sandstone cut contains a planar bedding joint dipping α = 38° (length L = 55 ft) and a vertical tension crack behind the crest. After a storm the tension crack fills to hw = 18 ft with water that drains along the joint to the face. Block weight W = 210,000 lb/ft. Joint c = 300 psf, φ = 32°. No seismic load. Rock RMR = 58 (fair).

Most nearly, the factor of safety with the crack full of water is:

A. 0.93
B. 0.85
C. 0.78
D. 0.70

Given equations / data

> FS = [cL + (W cos α − PWR − PWJ sin α) tan φ]/(W sin α + PWJ cos α); PWJ = ½γw hw²; PWR = ½γw hw L

Worked solution

1. PWJ = ½(62.4)(18)² = 10,109 lb/ft
2. PWR = ½(62.4)(18)(55) = 30,888 lb/ft
3. N′ = 165,482 − 30,888 − 6,224 = 128,371
4. Resisting = 16,500 + N′ tan32 = 96,715
5. Driving = 129,289 + 7,966 = 137,255
6. FS = 0.70 (dry 0.93)

Final answer

0.70

Why the other values are wrong

- **0.93** — ignored water forces
- **0.85** — omitted uplift PWR on the joint
- **0.78** — omitted the tension-crack force PWJ

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.3 Rock Slope Failure, p. 104

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=8cX88YWUo4U, https://www.youtube.com/watch?v=jvEQeZuqydI (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-33 · Galvanized coating life

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

MSE wall steel strips are hot-dip galvanized with a 86 μm zinc coating. Backfill resistivity tests average 4,200 Ω·cm, pH 7.1, chlorides 60 ppm. Estimate how long the zinc lasts using the AVERAGE galvanized loss constants before the base steel starts to corrode.

Most nearly, the zinc coating life is:

A. 7 years
B. 2 years
C. 5 years
D. 3 years

Given equations / data

> $x = K\,t^{n}$; galvanized: $n = 0.65$, $K = 25$ (avg) / 50 (max) μm

Worked solution

1. 4,200 Ω·cm → moderately corrosive (2,000–5,000)
2. t = (x/K)^(1/n) = (86/25)^(1/0.65) = 3.44^1.538 = 6.7 yr

Final answer

7 years

Why the other values are wrong

- **2 years** — used the maximum K = 50
- **5 years** — divided by n instead of taking the 1/n root
- **3 years** — used carbon-steel constants

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.17.1–3.17.2, pp. 197–198

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-34 · Collapsible soil settlement

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A 9 ft thick layer of dry loess will support a spread-footing warehouse. A double-oedometer test on a sample from mid-layer (initial e0 = 0.98) shows: under the design stress (dry) e = 0.94; after inundation at that same stress e = 0.83. Natural w = 6%, γd = 86 pcf. The owner asks how much additional settlement occurs if lawn irrigation wets the layer after construction.

Most nearly, the collapse settlement caused by wetting is:

A. 8.2 in.
B. 11.9 in.
C. 6.0 in.
D. 0.5 in.

Given equations / data

$$\Delta H = H\,\Delta e/(1 + e0)$$

Worked solution

1. Δe (collapse) $= 0.94 - 0.83 = 0.11$
2. $\Delta H = 9 \times 0.11/(1 + 0.98) = 0.500$ ft $= 6.0$ in.

Final answer

6.0 in.

Why the other values are wrong

- **8.2 in.** — included the dry load-compression as collapse
- **11.9 in.** — forgot to divide by (1 + e0)
- **0.5 in.** — left the answer in feet labeled inches

References (Jayce's library)

- Vandevelde, ASCE PE Review — Foundation Engineering (S = HΔe/(1+e0) worked example); NCEES PE Civil Reference Handbook v2.1 §3.2 Consolidation (strain form)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU, https://www.youtube.com/watch?v=M9uBSVNmUgU (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-35 · Secondary compression (organic)

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A 14 ft layer of organic silt (OH, organic content 14%) under a preloaded embankment reaches the end of primary consolidation at t = 1.8 years. The strain-based coefficient of secondary compression is Cα = 0.012. The design life of the roadway is 40 years from construction. Primary settlement was 22 in.

Most nearly, the secondary settlement from end of primary to end of design life is:

A. 3.2 in.
B. 6.3 in.
C. 1.0 in.
D. 2.7 in.

Given equations / data

> Ss = Cα H0 Δlog t (Cα in strain form)

Worked solution

1. Δlog t = log(40/1.8) = 1.3468
2. Ss = 0.012 × 14 × 1.3468 = 0.2263 ft = 2.7 in.

Final answer

2.7 in.

Why the other values are wrong

- **3.2 in.** — used log t2 (started the clock at 1 yr)
- **6.3 in.** — used natural log
- **1.0 in.** — divided by (1 + e0) again although Cα is in strain form

References (Jayce's library)

- Vandevelde, ASCE PE Review — Foundation Engineering (Ss = Cα H0 Δlog t)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=IGS94AWmPWA (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.